

Material and CMF Proposal: UnBROKEN Skate Wrap

This comprehensive proposal outlines the Material and Color, Material, and Finish (CMF) strategy for the **UnBROKEN Skate Wrap**. It is designed to bridge the gap between high-performance industrial protection and the raw, authentic aesthetics of the global and local skateboarding subculture.

2.1 Material Strategy

The material strategy for the **UnBROKEN Skate Wrap** is fundamentally guided by the principle of **"Authenticity through Durability"**. Within the skateboarding subculture, subcultural appropriation and spatial identity are tightly linked to an anti-establishment, DIY (Do-It-Yourself) ethos. Therefore, the choice of materials must intentionally avoid a "premium luxury" or overly corporate appearance, which would trigger subcultural rejection and label the product as a gadget for "posers".

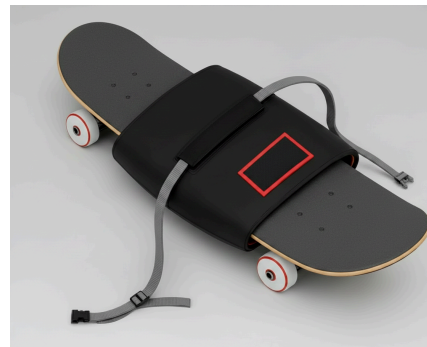
Instead, the strategy relies on a **Post-Modern Eclectic hybridization** : blending high-performance industrial protection textiles with raw, street-aligned materials. The materials are selected to function as a "legitimate tool" for the skater's daily routine, acting as a robust canvas that balances technological safety with high-edge personal identity and cultural expression.



2.2 Main Parts and Possible Materials

To ensure structural integrity under repetitive mechanical stress, the 85 cm Skate Wrap is mapped into functionally specialized components:

- **Main Outer Shell:** This is the primary sacrificial surface in direct contact with the skateboard's 80-grit abrasive *liner* (griptape). The materials under evaluation are ultra-high-performance synthetic fibers, specifically **Aramid 1414 (Kevlar®)** or **Ultra-High Molecular Weight Polyethylene (UHMWPE)** woven fabrics. These fibers provide the extreme cut and abrasion resistance required to withstand severe frictional wear.
- **Internal Organizer Tray:** The interior canvas that unfolds to expose the dedicated tool slots. It requires a balance of texturized durability and flexibility, utilizing either a heavy-duty **Stretch Denim** (blended with 2-3% Spandex) to mirror everyday streetwear textiles or a **Nylon fabric with a texturized matte finish**.
- **Transport Straps:** Engineered to support the full suspended weight of the board and accessories during movement. This component requires heavy-duty, high-tensile **Nylon webbing** structurally calculated for an ergonomic strap length of 64 cm.
- **Hardware (Buckles & Sliders):** High-impact **Polyoxymethylene (POM / Acetal)** industrial-grade side-release buckles and ladderlock buckles. These allow for quick-strap tensioning adjustments, high impact resistance if dropped on concrete, and fast manual release.
- **Airtight Tool Pockets:** Dedicated internal enclosures designed to store skate wax and bearing lubricants. To block fluid migration and prevent oils or melted wax from ruining adjacent streetwear or the backpack, these compartments require a thick internal polymer barrier coating of **Polyurethane (PU)** or eco-friendly **Thermoplastic Polyurethane (TPU)**.
- **Padding and Comfort:** Positioned on the internal back-facing surface to safeguard the user against the hard edges of the board, trucks, and hardware during commutes. It uses a core of high-density, closed-cell **EVA or Polyurethane Foam (busa)** for ergonomic weight distribution.
- **Aesthetic/Cultural Accents:** Structural inserts of traditional textiles such as **Batik or Tenun** patterns, paired with flexible **Rattan outfit trims**. These elements serve as a physical tribute to regional identity and national design heritage.



2.3 Material Properties Needed

Every material selected for the Skate Wrap must satisfy specific technical and subcultural requirements:

- **Extreme Abrasion Resistance (EN388 Level 4 Minimum):** The outer textile face must resist destructive friction caused by the sharp silicon carbide grains of skateboard griptape. It must comply with **EN388 mechanical risk guidelines** to guarantee the case will not wear through during daily friction cycles.
- **Cut, Piercing, and Scratch Resistance (ISO 13997):** High tensile modulus is mandatory to prevent sharp tools (screwdrivers, axle files) or the metal edges of skate trucks from piercing through the casing under sudden impacts.



- **Thermal Oxidative Degradation Resistance:**

Synthetic components (Nylon webbing, PU coatings) must be stabilized against UV-light exposure and urban radiant heat. This ensures the polymer chains do not cross-link or undergo embrittlement during outdoor street sessions or public transport commutes.

- **Hydrophobic and Oleophobic Barrier**

Performance: The wax-retention cell must display an absolute liquid barrier to organic solvents, greases, and low-temperature melting hydro-carbon waxes, keeping the contamination isolated.



- **Low Weight/High Strength Ratio:** While exact mass limits are to be defined, the core assembly must favor low bulk density. This minimizes dead weight, helping to reduce shoulder and spinal strain during multi-modal transport.
- **Elastic Elasticity & Flexural Memory:** The core structure must remain highly flexible, possessing the ability to tightly wrap around a standard deck or rapidly fold back into an ultra-compact pocket size when empty.



2.4 Possible Production Processes

The manufacturing of the Skate Wrap involves a mix of automated and artisanal techniques:

- **Heavy-Duty Sewing and Structural Stitching:** Joining the high-performance fabrics, Nylon webbing, and flexible layers via industrial computerized lockstitch machines. Specialized high-strength synthetic threads (such as bonded Nylon or Aramid threads) are required, utilizing bar-tack sewing patterns at high-stress strap anchor points.
- **Polymer Knife-Over-Roll Coating / Hot-Melt Lamination:** The application of liquid PU or TPU matrices directly onto the interior Nylon or Denim face. This creates a seamless, continuous polymer film that guarantees fluid-tight containment for lubricants and waxes.
- **Surface Modification via Photochemical UV Grafting:** For advanced configurations using UHMWPE fibers, high-energy UV light surface treatment can be utilized. This modifies the chemically inert surface of the polyethylene fibers, improving mechanical interlocking and adhesion with backing coatings or adjacent structural adhesives.
- **Artisanal Rattan Filtration Processing:** A specialized manual bending and weaving technique to process raw rattan strips into flexible, low-profile framing borders. This anchors the traditional aesthetic accent directly into the modern textile assembly.

2.5 CMF Palette Direction

The CMF direction is defined by the moodboard "**Identity, edge & expression,**" focusing on a **Post-Modern Eclectic** style.

Colors:

- *Base:* **Industrial Carbon Black** and **Asphalt Gray**. These choices emphasize extreme durability, anchor the product in an urban architectural context, and effectively hide grease, street soot, and griptape debris.
- *Accent:* **Signature Kinetic Orange**. A high-visibility, high-contrast hue used strategically on branding labels, internal stitching, or pocket trims to convey energy, emergency tools, and raw edge.
- *Identity:* **Earth-toned Organic Browns** derived natively from the Rattan frames and the vegetable dyes of the localized Batik/Tenun prints.



Materials & Textures: A stark, deliberate contrast between "cold/sterile" high-tech protection textiles (the smooth, micro-grid weaves of Nylon and Aramid) and the "warm/tactile" organic roughness of raw denim, natural rattan fiber, and printed cotton warp yarns.



Finishes: Strictly industrial **Matte, Raw, and Rugged finishes**. Glossy treatments or delicate luxury coatings are rejected; the surfaces must look functional, highly tactile, and completely street-ready.

2.6 Material Risks

- **Cost Escalation vs. DIY Target Market:** Integrating specialized technical fibers like Kevlar® or UHMWPE exponentially increases raw material costs per unit. If production costs push the MSRP into a "luxury" tier, price-sensitive skaters will abandon the product and return to improvised, zero-cost solutions (like old socks or standard plastic bags).
- **Chemical Vulnerabilities in Low-Cost Coatings:** If standard PVC coatings with plasticizers like DOP are utilized to reduce costs, they face long-term chemical risks, including plasticizer migration when exposed to skate wax oils, leading to sticky or degraded surfaces.
- **Subcultural Rejection due to Aesthetics:** If the synthetic shell looks overly engineered, neon-colored, or resembling orthopedic gear, it risks being labeled as a "poser gadget". The visual design must remain understated and culturally integrated.



- **Abrasive Fatigue and Wear Life Cycle:** No fabric is entirely immune to the continuous grinding action of a silicon carbide skateboard line. The fiber matrix will eventually experience micro-fraying, meaning exact wear life-cycles must undergo strict abrasion field-testing.

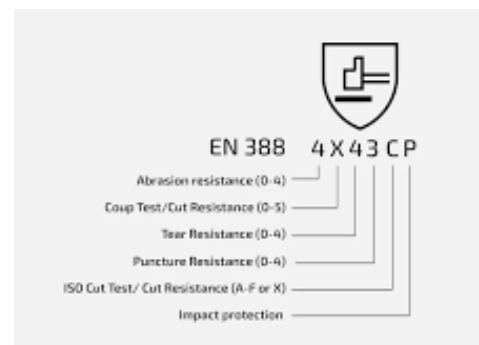
2.7 Alternative Materials

- **Alternative for Outer Shell (Cost Reduction):** Heavyweight **Stretch Denim (14oz+)** or ultra-dense **1680D Ballistic Nylon** can replace Aramids for areas not directly exposed to severe friction, maintaining a familiar streetwear look while lowering material costs.
- **Alternative Eco-Friendly Coatings: Thermoplastic Polyurethane (TPU) or Ethylene-Vinyl Acetate (EVA)** films are ideal substitutes to replace toxic PVC layers, offering excellent oil resistance and waterproof properties without hazardous plasticizers.
- **Alternative Structural Stabilizers:** If raw rattan processing faces supply chain limits, heavy-weight **Vegetable-Tanned Leather** scrap pieces or industrial-grade **Textile Interlining (Kain Keras)** can supply the internal stiffness needed to keep the wrap's silhouette crisp.

2.8 How to Explain Materials in the Technical Report

For an industrial manufacturer or OEM partner, the description must be purely quantitative, data-driven, and focused on regulatory compliance:

- **Regulatory Compliance:** Explicitly document compliance with **EN388 Standards** for Mechanical Risks (targeting a Level 4 rating for abrasion and cut resistance) and **EN420 General Safety Protocols**.
- **Polymer Chemistry Details:** Specify the exact texturized denier of the Nylon backing yarns and clarify the high-modulus molecular characteristics of the chosen Aramid/UHMWPE weaves.
- **Technical Blueprints:** Include detailed engineering assets, specifically **isometric views, exploded assembly diagrams, and cross-sections**. These must map out the exact placement of internal layers, such as the PU/TPU fluid-tight barriers versus the foam padding placement.



2.9 How to Communicate Materials in the Visual/Commercial Report

For investors, retailers, and end-consumers, the material communication must shift entirely toward lifestyle, identity, and real-world performance:

- **Action-In-Motion Context:** Avoid clean, sterile studio settings. Utilize rugged lifestyle photography showcasing the Skate Wrap in authentic, real-world urban scenarios—such

as mid-session street repairs, multi-modal public transit commutes, or encounters with concrete ledges.

- **The Personalization Component:** Frame the textiles not just as protective shields, but as an interactive tool for self-expression. Highlight the **online digital configurator**, showcasing how users can select custom color combinations and culturally rich pattern inserts (Batik/Tenun) to match their personal style.
- **Subcultural Validation:** Center the messaging around legitimacy, backing the design with active endorsements from respected professional skaters (e.g., **Max Barrera, Itzel Granados**), ensuring the material durability is verified by those who ride daily.

2.10

Part / Component	Proposed Material	Key Properties	Process	Finish / Color	Risk / Alternative
Main Body / Outer Shell	Cut-resistant Nylon reinforced with Aramid (Kevlar) or UHMWPE	Durable, light, safe; extreme abrasion resistance (EN388 standards)	Manual and machine sewing; possible RF welding or UV grafting for fiber modification	Matte / textured; neutral base (Industrial Black/Gray)	Needs testing for Iija (griptape) resistance; alternative: Rattan fitrit for an ethnic-modern eclectic look
Strap / Support	Reinforced Nylon webbing	Flexible, strong, comfortable; high tensile strength to support board weight	Cutting + sewing	Textile finish; neutral with vibrant brand accents (Orange)	Check resistance; ensure safety during high-speed commuting
Connector / Joint	Industrial-grade side-release and ladderlock buckles	Precise, repeatable, strong; allows for one-click access	Injection / CNC / Commercial sourcing	Neutral / technical finish	Check tolerance; ensure hardware survives urban impacts
User Touch Area (Back Panel)	Foam (busa) or padded technical textile	Soft, washable, ergonomic; provides a buffer against the wooden board	Foam / textile / rubber assembly	Grip texture; comfortable, breathable finish	Check cleaning; rattan frames may cause discomfort if unpadded
Replaceable Module (Internal Tray)	Stretch Denim (denim/spandex blend) or elastic loops	Modular, easy to remove, flexible for various tool sizes	Assembly / fastener	Brand color or Batik/Tenun patterns for subcultural identity	Check wear from metal tool friction
Airtight Pockets	PU (Polyurethane) or PVC coated fabric	Leak-proof, airtight, and chemical/oil resistant	Dipping / coating / RF welding	Sealed/technical look	Environmental risk with PVC; alternative: TPU or EVA for eco-friendly waterproofing

Status of Specifications:

- **Exact Material Weight:** To be defined.
- **Target MSRP:** To be defined.
- **PVC/EVA Welding Compatibility:** Needs testing.
- **Final Abrasion Life Cycle:** Needs testing.